Stochastic Processes 2021/22 Semester 1

Tutorial 8

1. A system moves between 3 states as a (continuous time) homogeneous Markov process with transition rate matrix

$$Q = \begin{pmatrix} -2\lambda & \lambda & \lambda \\ \lambda & -2\lambda & \lambda \\ \lambda & \lambda & -2\lambda \end{pmatrix}$$

   (a) Conditional on the process starting in any state i at time 0, verify by checking the forward differential equations and the appropriate initial conditions that the probability that it is in state j at time t is given by

$$p_{ij}(t) = \begin{cases} \frac{1}{3} + \frac{2}{3}\exp(-3\lambda t) & \text{if } j = i \\ \frac{1}{3} - \frac{1}{3}\exp(-3\lambda t) & \text{if } j \neq i. \end{cases}$$

   (b) Now prove these formulae from first principles.

   (c) Find the stationary distribution $\boldsymbol{\pi}$ of the process.

   (d) Verify that

$$\lim_{t\to\infty} p_{ij}(t) = \pi_j$$

   for all $i, j \in S$.

2. A very simple "healthy-sick-dead" model is given by a (continuous time) homogeneous Markov process with state space $\{H, S, D\}$ and transition rate matrix

$$Q = \begin{pmatrix} -(\lambda+\nu) & \lambda & \nu \\ \mu & -(\mu+\nu) & \nu \\ 0 & 0 & 0 \end{pmatrix}$$

   (a) Verify that

$$\begin{aligned} p_{HH}(t) &= \left(\frac{\mu}{\lambda+\mu} + \frac{\lambda}{\lambda+\mu}e^{-(\lambda+\mu)t}\right)e^{-\nu t} \\ p_{HS}(t) &= \left(\frac{\lambda}{\lambda+\mu} - \frac{\lambda}{\lambda+\mu}e^{-(\lambda+\mu)t}\right)e^{-\nu t} \\ p_{HD}(t) &= 1 - e^{-\nu t}. \end{aligned}$$

   satisfy the forward differential equations and the appropriate initial conditions

   (b) Write down the remaining probability functions.

   (c) Prove the formulae given in subquestion (a) from first principles.

   (d) Write down $\lim_{t\to\infty} P(t)$, where $P(t)$ is the stochastic matrix of transition probabilities associated with the time t.

   (e) Find the transition matrix of the associated jumping chain, and hence give an alternative derivation of the limiting behaviour of the process.

3. A continuous time Markov process on the state space $S = \{1, 2, 3, 4, 5, 6\}$ has transition rate matrix given by

$$Q = \begin{pmatrix} -5 & 1 & 1 & 1 & 1 & 1 \\ 0 & -4 & 1 & 3 & 0 & 0 \\ 2 & 0 & -4 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -2 & 2 \\ 0 & 0 & 0 & 0 & 3 & -3 \end{pmatrix}$$

   (a) Write down the transition matrix of the corresponding jumping chain.

   (b) Hence classify the states of the process into open and closed classes, identifying also transience/recurrence.

   (c) Find the absorption probabilities associated with each of the closed classes.

4. A system moves between 3 states as a (continuous time) Markov process with matrix of transition rates (generator)

$$Q = \begin{pmatrix} -3 & 1 & 2 \\ 3 & -4 & 1 \\ 4 & 2 & -6 \end{pmatrix}$$

   Observe that all states form a single closed class. Find the stationary distribution of the process.

5. Consider the general birth-and-death process $\{(X(t)\}_{t\geq 0}$ on the state space $S = \{0, 1, 2, \ldots\}$. This has matrix of transition rates (generator) $Q = (q_{ij})$ given by

$$\begin{aligned} q_{i,i+1} &= \lambda_i, && i \geq 0 && \text{(birth rate in state } i\text{)}, \\ q_{i,i-1} &= \mu_i, && i \geq 1 && \text{(death rate in state } i\text{)}, \\ q_{ii} &= -(\lambda_i + \mu_i) && i \geq 1 \\ q_{00} &= -\lambda_0 \\ q_{ij} &= 0 && \text{for all other } j \neq i. \end{aligned}$$

   Note that, for all i, $q_{ii} = -\sum_{j\neq i} q_{ij}$ as usual.

   (a) Suppose first that $\lambda_i = \lambda > 0$ for all $i \geq 0$ and that $\mu_i = \mu > 0$ for all $i \geq 1$ (a continuous time random walk with 0 as a reflecting barrier).

      i. Verify that the equation $\boldsymbol{\pi} Q = \mathbf{0}$ has the solution

$$\pi_i = k \left(\frac{\lambda}{\mu}\right)^i, \qquad i \geq 0,$$

      for any $k > 0$.

      ii. Hence show that a stationary distribution for the process $\{(X(t)\}_{t\geq 0}$ exists if and only if $\lambda < \mu$, and in this case determine it.

   (b) Now consider the *linear* birth-and-death process, in which $\lambda_i = \lambda i$ for all $i \geq 0$ (where $\lambda > 0$) and $\mu_i = \mu i$ for all $i \geq 1$ (where $\mu > 0$)—so that λ can be thought of as a birth rate per individual and μ can be thought of as a death rate per individual. By considering the associated jumping chain, show that here 0 is an absorbing state for the process and (by reference to the theory of discrete time random walks) find the associated vector of absorption probabilities.

6. Consider the following set-up:

   - There are M machines operating in a factory.
   - *Each* machine working at time t has probability $\mu h + o(h)$ of breaking down in the interval $(t, t+h)$—independently of everything prior to time t.
   - A technician works on any machines that are broken. If there is at least one machine broken at time t, then the probability that exactly one machine will be repaired in the interval $(t, t+h)$ is $\lambda h + o(h)$—again independently of everything prior to time t and of the breakdown process.
   - Let $X(t)$ be the number of machines working at time t.

   (a) Write down the transition rate matrix Q for the Markov process $\{X(t)\}_{t\geq 0}$.

   (b) Verify that the stationary distribution $\boldsymbol{\pi}$ for the process is given by

   $$\pi_n = k\frac{\left(\frac{\lambda}{\mu}\right)^n}{n!}$$

   for a constant k which you should identify.

   (c) To which distribution does the stationary distribution $\boldsymbol{\pi}$ (of the number of working machines) converge as $M \to \infty$?